MIL-P-50578C (AR)
AMENDMENT 4
06 August 1982

SUPERSEDING
AMENDMENT 3
04 November 1981

MILITARY SPECIFICATION

PROJECTILE, 155MM, HE, RA, M549 & M549A1
(WITH WARHEAD AND MOTOR BODY)
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-P-50578C (AR), dated 20 June 1979 and is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 4

* Add 3.3.6 and 3.3.7 as follows:

"3.3.6 Metal parts defects. There shall be no cracks, pipes or thin wall sections in the warhead.

3.3.7 Foreign material. There shall be no foreign material in the explosive charge."

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4.4.2.2, Minor 203: Delete in its entirety and substitute the following:

"Minor 203 Anti-seize compound not 360 degrees around liner or on its base. 100% 3.2 Visual".

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4.4.2.4, Minor 205: Delete in its entirety and substitute the following:

"Minor 205 Anti-seize compound missing from threads. 0.65% 3.2 Visual."

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* Add 4.4.3.1.2 and 4.4.3.1.3 as follows:

"4.4.3.1.2 Metal parts defects (Critical defect). All radiographs of warhead section loading assemblies shall be examined for evidence of cracks, pipes and thin wall sections in the warhead. The presence of any such indication shall cause that assembly to be classed defectiver and removed from the lot. (See Section 6.14).

4.4.3.1.3 Foreign material (Critical defect). All radiographs of warhead section loading assemblies shall be examined for foreign material in the explosive charge. The presence of any foreign material shall cause that assembly to be classed defective and removed from the lot."

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* 4.5.1.1, delete the following phrase:

"unless identified as extraneous artifacts or metal parts irregularities" and substitute the following:

"unless identified as film or processing related or as acceptable metal parts irregularities."

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* Add the following new paragraph:

"4.5.1.5 Operator qualifications. For operation of radiograph equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of radiographic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II."

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4.5.3.1, 4.5.3.2: Delete in their entirety and substitute the following:

"4.5.3.1 Functioning. All test projectiles shall be conditioned and fired at 70 degrees Fahrenheit (^{O}F) at Charge 8, M203 series propelling charge conditioned at 70°F. In order to minimize the effects of meteorological conditions the

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test shall be fired in either one (for normal inspection) or three (for tightened inspection) series of 10 rounds each. The first two subgroups for tightened inspection shall be fired at a quadrant elevation of 33 degrees and the last subgroups for tightened inspection or the single group for normal inspection shall be fired at a quadrant elevation of 51 degrees. Time lapses of more than 15 minutes between consecutive rounds shall not be permitted and it is desirable that each 10 round series be completed within one (1) hour. Range probable errors shall be calculated for each 10 round subgroup and then a pooled range probable error (for tightened inspection) shall be computed for determination of compliance with 3.6.1.2.1.

$$\text{Subgroup RPE (percent)} = RPEj = \frac{X_i - 1/n\ (X)^2}{n-1} \times \frac{100\% \times .6745}{\text{Ave. Range (j)}}$$

$$\text{Pooled RPE (percent)} = RPEp = \frac{RPE_1^2\ ^2(n_1-1) + RPE_2^2\ (n_2-1) + RPE_3^2\ (n_3-1)}{DF}$$

Where X = Individual horizontal range within each subgroup.

Where DF = Degrees of Freedom = n + n ... n(j) - number of subgroups.

4.5.3.2 Safety. The projectiles for this test shall be divided into three equal groups. Group 1 shall be conditioned and fired at -65°F at Charge 8, M203 series propelling charge conditioned at -65°F. Group 2 shall be conditioned and fired at 145°F with an M203 propelling charge conditioned at 145°F. Group 3 shall be conditioned at 145°F and fired with a PXR-6297 proof propelling charge conditioned at 145°F. The cold phase (Group 1) shall be fired at a quadrant elevation of 51 degrees and the hot phases (Group 2 and 3) shall be fired at a quadrant elevation of 24 degrees."

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Add the following new paragraph:

"6.13 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreement QSTAG 605. When amendment, revision or cancellation of this specification is proposed which will affect or violate the international

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agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices, if required."

* Add the following new paragraph:

"6.14 The procuring activity will be notified of any projectile rejected for metal parts defects found upon inspection of the radiographs. The rejected projectile will be held pending disposition by the procuring activity."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
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